

Lab Manual for Digital Signal Processing



Lab Manual for
Digital Signal Processing lab (CSE-416)
Credit: 1, Contact hour: 2 Hours Per week



Department of Computer Science and Engineering, Varendra
University, Rajshahi, Bangladesh

Lab Instructions:

- Students should come with thorough preparation for the experiment to be conducted.
- Students will not be permitted to attend the laboratory unless they bring the practical record fully completed in all respects pertaining to the experiment conducted in the previous class.
- Be honest in developing and representing your program. If a particular program output appears wrong repeat the program carefully.
- Strictly observe the instructions given by the Faculty / Lab. Instructor.
- Take permission before entering the lab and keep your belongings in the racks.
- NO FOOD, DRINK, IN ANY FORM is allowed in the lab.
- TURN OFF CELL PHONES! If you need to use it, please keep it in bags.
- Avoid all horseplay in the laboratory. Do not misbehave in the computer laboratory.
Work quietly.
- Save often and keep your files organized
- Do not reconfigure the cabling/equipment without prior permission.
- Do not play games on systems.
- If you finish early, spend the remaining time to complete the laboratory report writing.
Come equipped with calculator and other materials related to lab works.
- Handle instruments with care. Report any breakage or faulty equipment to the Instructor.
Shutdown your computer you have used for the purpose of your experiment before leaving the Laboratory.
- Violation of the above rules and etiquette guidelines will result in disciplinary action.

Varendra University

COURSE SYLLABUS

1	Faculty	Faculty of Science & Engineering
2	Department	Department of CSE
3	Program	B.Sc. in Computer Science and Engineering
4	Name of Course	Digital Signal Processing Lab
5	Course Code	CSE-416
6	Trimester and Year	Fall-2020
7	Pre-requisites	CSE-112, CSE-233
8	Status	Core Course
9	Credit Hours	1
10	Section	
11	Class Hours	3 hours
12	Class Location	
13	Name (s) of Academic staff / Instructor(s)	
14	Contact	
15	Counseling Hours	
16	Text Book	J.G. Proakis and D.G. Manolakis, Digital Signal Processing 3rd edition, Prentice-Hall
17	Reference	R.G Lyons, Understanding Digital Signal processing, 2nd edition, Prentice-Hall. (Amazon's top-selling for five straight years)
18	Equipment & Aids	Lab Sheet Text Book MATLAB & Python
19	Course Description	The lab complements the CSE-416 Digital Signal Processing (DSP) course by providing experience designing and understanding real digital signal processing systems. This class involves a series of weekly labs (some multi-week) that connect DSP concepts learned in the class with practical experiments using MATLAB software. Each lab culminates in a detailed write-up. These lab reports are the major assignments in this course.

Course Objectives	This course will help students to achieve the following objectives: 1. Design and implement a DSP system using MATLAB 2. Analyze and describe the functionality of a real-world DSP system 3. Work in teams to plan and execute the creation of a complex DSP system 4. Apply DSP system design to real world applications
Learning Outcomes	Upon completion of this course, students will be able to analyze Digital Signal Processing.
Teaching Methods	Lecture, Video Demonstration, Problem Solving, Project Development

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EXPERIMENT-1

Aim : INTRODUCTION ABOUT MATLAB

MATLAB® is a high-performance language for technical computing. It integrates computation, visualization, and programming in an easy-to-use environment where problems and solutions are expressed in familiar mathematical notation.

Typical uses include:

- Math and computation
- Algorithm development
- Data acquisition
- Modeling, simulation and prototyping
- Data analysis, exploration, and visualization
- Scientific and engineering graphics
- Application development, including graphical user interface building

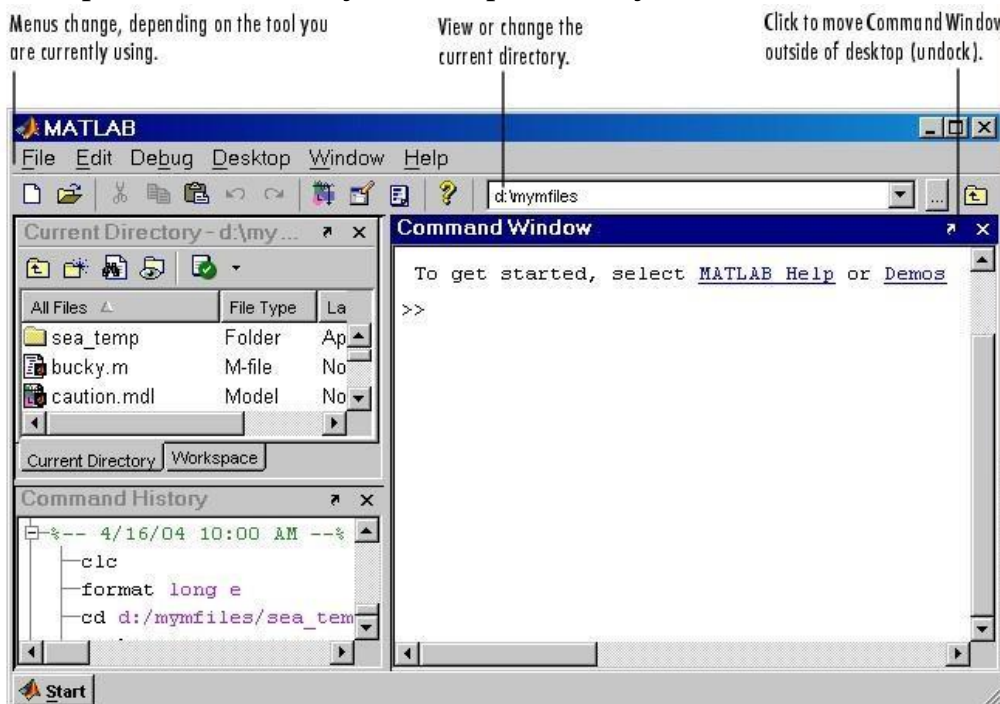
MATLAB is an interactive system whose basic data element is an array that does not require dimensioning. This allows you to solve many technical computing problems, especially those with matrix and vector formulations, in a fraction of the time it would take to write a program in a scalar no interactive language such as C or Fortran.

The name MATLAB stands for matrix laboratory. MATLAB was originally written to provide easy access to matrix software developed by the LINPACK and EISPACK projects. Today, MATLAB engines incorporate the LAPACK and BLAS libraries, embedding the state of the art in software for matrix computation.

Desktop Overview

Use desktop tools to manage your work in MATLAB. You can also use MATLAB functions to perform the equivalent of most of the features found in the desktop tools.

The following illustration shows the default configuration of the MATLAB desktop. You can modify the setup to meet your needs.



The MATLAB System

The MATLAB system consists of five main parts:

Desktop Tools and Development Environment. This is the set of tools and facilities that help you use MATLAB functions and files. Many of these tools are graphical user interfaces. It includes the MATLAB desktop and Command Window, a command history, an editor and debugger, and browsers for viewing help, the workspace, files, and the search path.

The MATLAB Mathematical Function Library. This is a vast collection of computational algorithms ranging from elementary functions, like sum, sine, cosine, and complex arithmetic, to more sophisticated functions like matrix inverse, matrix Eigen values, Bessel functions, and fast Fourier transforms.

The MATLAB Language. This is a high-level matrix/array language with control flow statements, functions, data structures, input/output, and object-oriented programming features. It allows both "programming in the small" to rapidly create quick and dirty throw-away programs, and "programming in the large" to create large and complex application programs.

Graphics. MATLAB has extensive facilities for displaying vectors and matrices as graphs, as well as annotating and printing these graphs. It includes high-level functions for two-dimensional and three-dimensional data visualization, image processing, animation, and presentation graphics. It also includes low-level functions that allow you to fully customize the appearance of graphics as well as to build complete graphical user interfaces on your MATLAB applications.

The MATLAB External Interfaces/API. This is a library that allows you to write C and Fortran programs that interact with MATLAB. It includes facilities for calling routines from MATLAB (dynamic linking), calling MATLAB as a computational engine, and for reading and writing MATfiles.

Programming:

Flow Control:

MATLAB has several flow control constructs:

- if, else, and elseif
- switch and case
- for while
- continue
- break
- try - catch
- return

Several special functions provide values of useful constants:

<u>pi</u>	3.14159265...
<u>i</u>	Imaginary unit, $\sqrt{-1}$
<u>j</u>	Same as i
<u>eps</u>	Floating-point relative precision, $\varepsilon = 2^{-52}$
<u>realmin</u>	Smallest floating-point number, 2^{-1022}
<u>realmax</u>	Largest floating-point number, $(2 - \varepsilon)2^{1023}$
<u>Inf</u>	Infinity
<u>NaN</u>	Not-a-number

Operators

Expressions use familiar arithmetic operators and precedence rules.

+	Addition
-	Subtraction
*	Multiplication
/	Division
\	Left division (described in "Matrices and Linear Algebra" in the MATLAB documentation)
^	Power
'	Complex conjugate transpose
()	Specify evaluation order

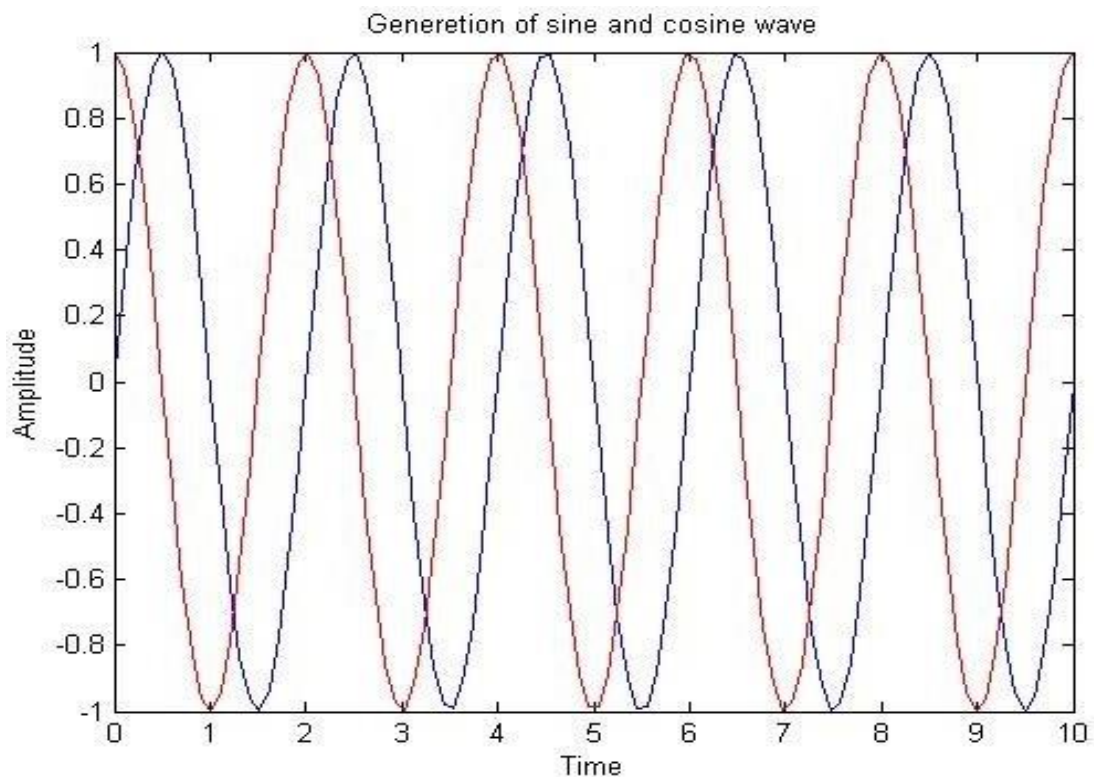
EXPERIMENT-2

AIM:- To develop sine and cosine function waveforms.

MATLAB Code :

```
t=linspace(0,10,100); x=sin(pi*t);  
plot(t,x);  
  
y=cos(pi*t); hold  
on  
plot(t,y,'red'); xlabel('Time');  
ylabel('Amplitude');  
title ('Generetion of sine and cosine wave');
```

Waveform :



EXPERIMENT-3

Aim : To develop elementary signals: Unit step signal, unit ramp signal, unit impulse signal, exponential signal.

MATLAB Code :

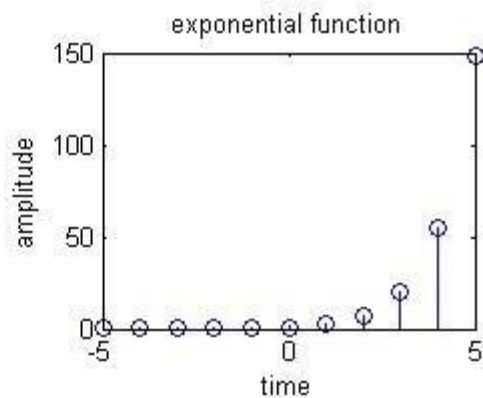
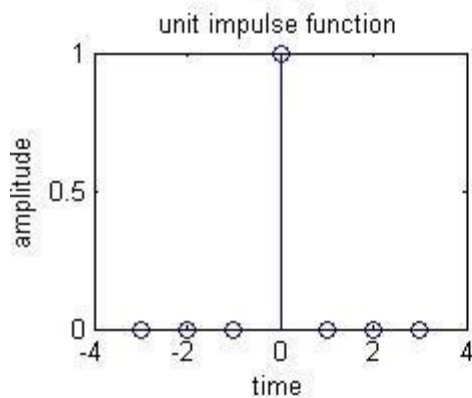
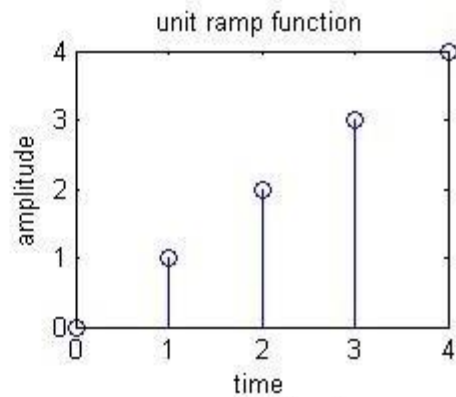
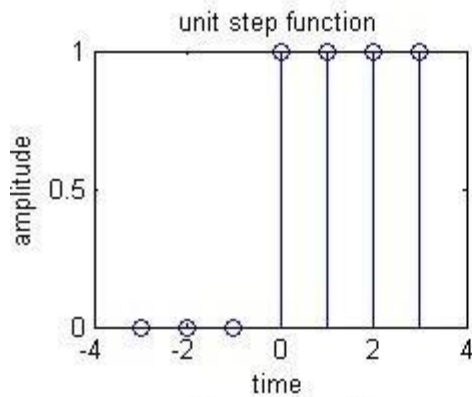
```
t=-3:1:3;  
y=[zeros(1,3),ones(1,4)];  
subplot(2,2,1); stem(t,y);  
xlabel('time');  
ylabel('amplitude');  
title('unit step function');
```

```
t1=0:1:4; y1=t1;  
subplot(2,2,2);  
stem(t1,y1);  
xlabel('time');  
ylabel('amplitude');  
title('unit ramp function');
```

```
t2=-3:1:3;  
y2=[zeros(1,3),1,zeros(1,3)];  
subplot(2,2,3); stem(t2,y2);  
xlabel('time');  
ylabel('amplitude');  
title('unit impulse function');
```

```
t3=-5:1:5; y3=exp(t3);  
subplot(2,2,4); stem(t3,y3);  
xlabel('time');  
ylabel('amplitude');  
title('exponential function');
```

Waveforms :



EXPERIMENT-4

Aim :To Develop program modules based on operation on sequences like:
Signal addition, Signal multiplication, Signal folding, Signal Shifting.

MATLAB Code :

Signal Addition:-

```
t=0:1:5;  
x=[1,2,3,4,5,6];  
subplot(4,3,1);  
stem(t,x); xlabel('time');  
ylabel('amplitude');  
title('x sequence');
```

```
t=0:1:5; y=[0,-  
1,1,2,3,2];  
subplot(4,3,2);  
stem(t,y); xlabel('time');  
ylabel('amplitude');  
title('y sequence');
```

```
z=x+y; subplot(4,3,3)  
stem(t,z);  
xlabel('time');  
ylabel('amplitude');  
title('addition of x and y sequence');
```

Signal Multiplication:-

```
t2=0:1:5;  
x1=[1,2,3,4,5,6];  
subplot(4,3,4);  
stem(t2,x1);  
xlabel('time');  
ylabel('amplitude'); title('x  
sequence'); t2=0:1:5;  
y1=[1,1,2,2,3,5];  
subplot(4,3,5);  
stem(t2,y1);  
xlabel('time');  
ylabel('amplitude');  
title('x sequence');
```

```
z1=x1.*y1; subplot(4,3,6)
stem(t2,z1);
xlabel('time');
ylabel('amplitude');
title('multiplication of x and y sequence');
```

Signal Folding:-

```
t4=0:1:5; x2=[1,2,3,4,5,6];
subplot(4,3,7);
stem(t4,x2); xlabel('time');
ylabel('amplitude');
title('x sequence');
```

```
z2=-x2; subplot(4,3,8);
stem(t4,z2);
xlabel('time');
ylabel('amplitude');
title('folding of x sequence');
```

```
t5=0:1:5; t5=-t4;
subplot(4,3,9); stem(t5,z2);
xlabel('time');
ylabel('amplitude');
title('folding of x sequence');
```

Signal Shifting:-

```
t6=0:1:5; x3=[1,2,3,4,5,6];
subplot(4,3,10);
stem(t6,x3); xlabel('time');
ylabel('amplitude');
title('x sequence');
```

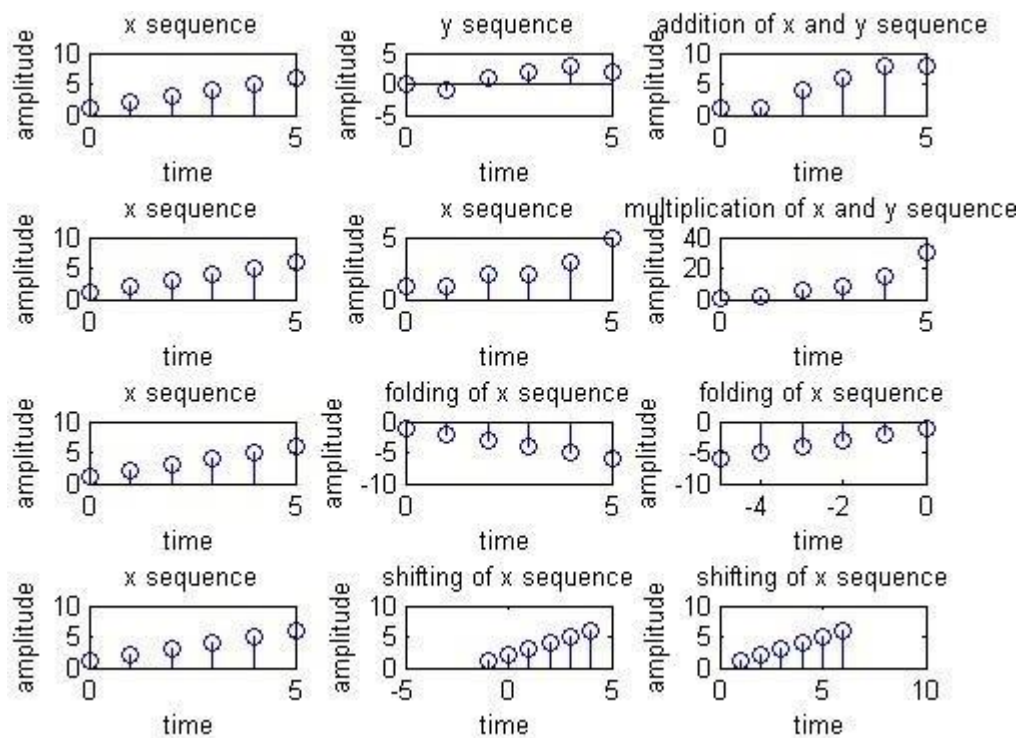
```
t7=0:1:6; t7=t6-1;
subplot(4,3,11); stem(t7,x3);
xlabel('time');
ylabel('amplitude');
title('shifting of x sequence');
```



```

t8=0:1:6; t8=t6+1;
subplot(4,3,12);
stem(t8,x3);
xlabel('time');
ylabel('amplitude');
title('shifting of x sequence'); Waveforms :

```



EXPERIMENT-5

Aim: To develop program to perform linear convolution of two input Sequence $x(n)$ and $h(n)$. Plot $x(n)$ and $h(n)$ and result of linear convolution as $y(n)$ on single plot and verify the result mathematically.

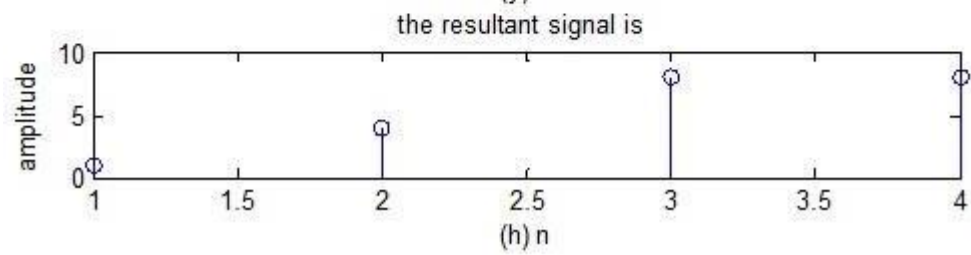
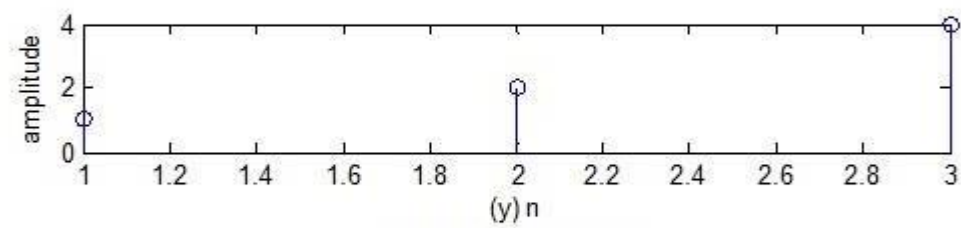
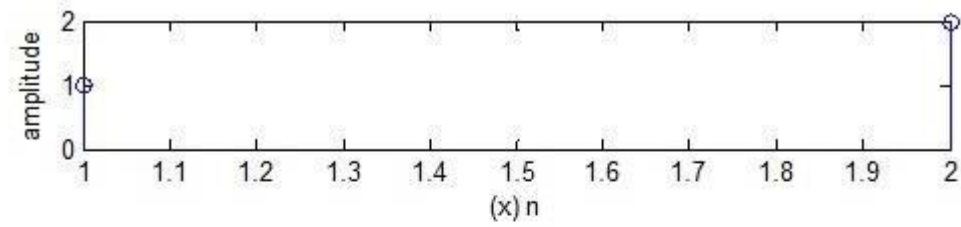
MATLAB Code :

```
x=input('enter the 1st sequence');
h=input('enter the 2nd sequence');
y=conv(x,h); subplot(3,1,1);
stem(x); xlabel('(x) n');
ylabel('amplitude');
subplot(3,1,2); stem(h);
xlabel('(y) n');
ylabel('amplitude');
subplot(3,1,3); stem(y);
xlabel('(h) n');
ylabel('amplitude');
title('the resultant signal is ');
```

Result on command window:

```
enter the 1st sequence[1 2]
enter the 2nd sequence[1 2 4]
```

Waveforms :



EXPERIMENT-6

Aim: To obtain the impulse response of the system described by difference equation. $y(n)-0.5y(n-1)=x(n)$.

MATLAB Code :

```
n=0:1:9
x=[ones(1,1),zeros(1,9)]
a=[1,-0.5]; b=[1];
h=filter(b,a,x)
stem(n,h);
xlabel('time');
ylabel('amplitude');
title('impulse response using difference equation');
```

n =

0 1 2 3 4 5 6 7 8 9

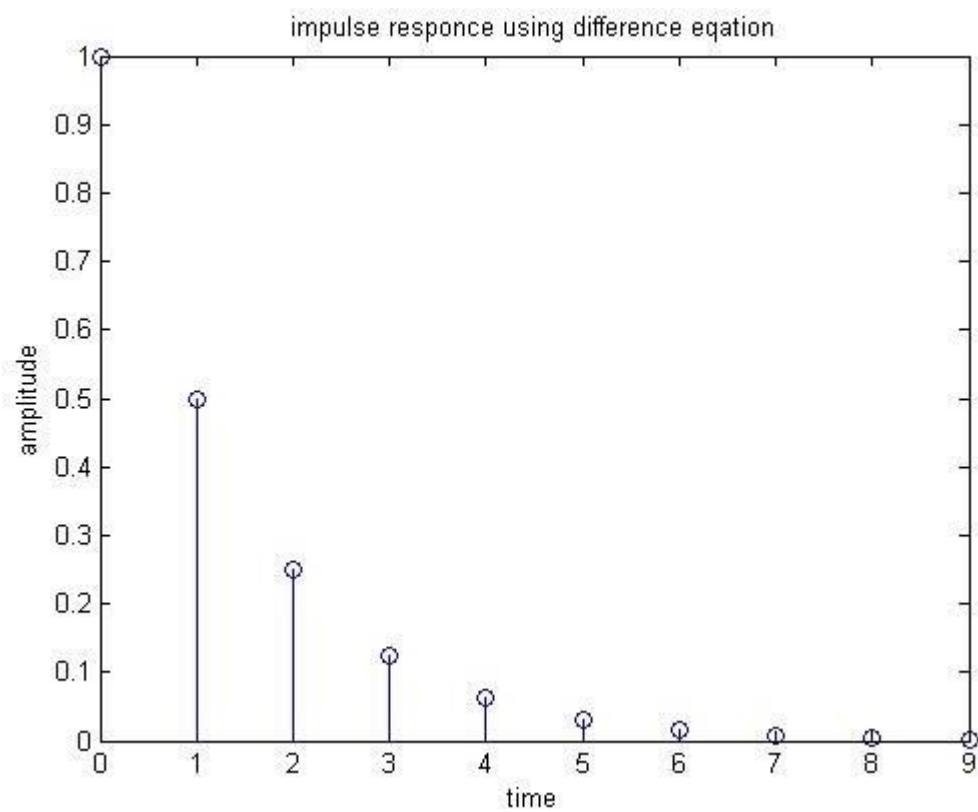
x =

1 0 0 0 0 0 0 0 0 0

h =

1.0000 0.5000 0.2500 0.1250 0.0625 0.0313 0.0156 0.0078 0.0039
0.0020

Waveforms :



EXPERIMENT-7

Aim :-To develop program for computing Z-Transform and inverse Ztransform.

MATLAB Code :

Z-Transform

```
syms z n
ztrans (2*2^n+4*(1/2)^n)
```

Result on command window:

```
ans = (2*z)/(z - 2) + (4*z)/(z - 1/2) inverse
```

Z-transform

```
syms z n
iztrans ((2*z)/(z - 2) + (4*z)/(z - 1/2))
```

Result on command window:

```
ans = 2*2^n + 4*(1/2)^n
```

EXPERIMENT-8

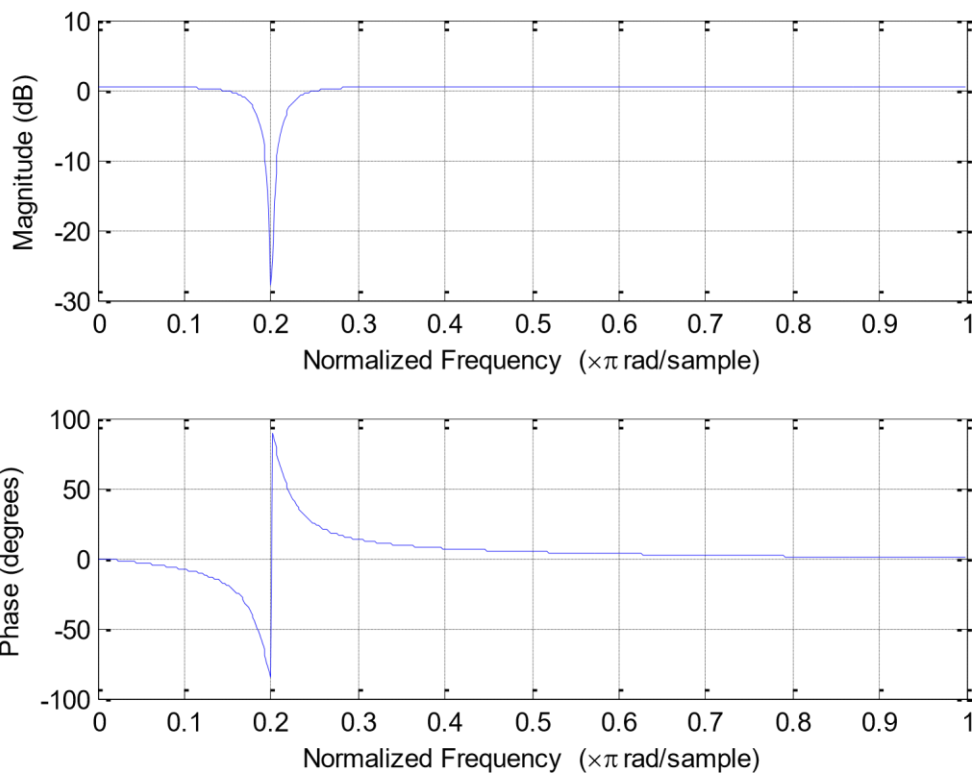
Aim : To develop program for finding magnitude and phase response of LTI system described by system function $H(z)$.

MATLAB Code :

$$H(z) = (1 - 1.6180z^{-1} + z^{-2}) / (1 - 1.5161z^{-1} + 0.878z^{-2})$$

```
b=[1 -1.6180 1]; a=[1  
-1.5161 0.878];  
freqz(b,a)
```

WAVEFORMS:



EXPERIMENT-9

Aim :- To develop program for computing DFT and IDFT using FFT.

MATLAB Code :

```
x=input ('enter the sequence ');  
n=8; x=fft(x,n)  
a=ifft(x)
```

Enter the sequence [1 0 1 0 1 0 1 0]

Result on command window:

```
x = 4    0    0    0    4    0    0  
0
```

Result on command window:

```
a = 1    0    1    0    1    0    1    0
```

EXPERIMENT-10

Aim :- To develop program for finding magnitude and phase response of LTI system described by difference equation:
 $y(n)=0.8y(n-2)+x(n)-x(n-2)$.

MATLAB Code :

```
clc; b=input('enter the coeff. of  
x(n)'); a=input('enter the coeff. of  
y(n)');  
N=200;  
[H,W]=freqz(b,a,N);  
subplot(2,1,1);  
absH=abs(H);  
angH=angle(H);  
plot(W,absH);  
xlabel('W'); ylabel('abs  
H'); title('LTI  
magnitude');  
subplot(2,1,2);  
plot(W,angH);  
xlabel('W');  
ylabel('angle H');  
title('LTI angle');
```

Result on command window:

```
enter the coeff. of x(n)[1 0 -1]  
enter the coeff. of y(n)[1 0 -0.81]
```


WAVEFORMS:

